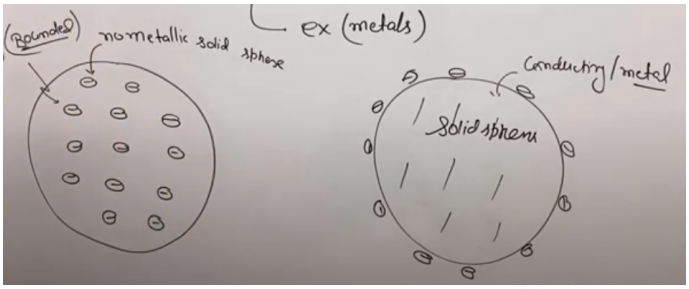
Conductors

Properties of Conductors (in Electrostatic condition):

→ Conductors have very large no. of free electrons.

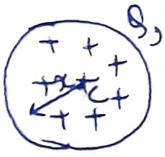


- → Not bounded
- → Free from attraction of nucleus
- → They can roam in lattice of conductor
- → They are responsible for transfer of charge and energy
- → Ex - Metals.

→ Any charge given to a conductor will reside at outer surface.

Non-Metal

① Solid sphere:



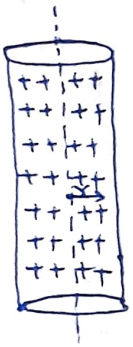
$\rightarrow E_{in} = \frac{KQr}{R^3} = \frac{\rho r}{3\varepsilon_0}$

$\rightarrow V_{in} = \frac{KQ}{2R^3}\{3R^2 - r^2\}$

$\rightarrow V_c = \frac{3KQ}{2R}$

$\rightarrow U_{self} = \frac{3KQ^2}{5R}$

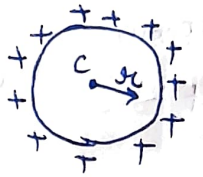
② Infinitely long solid cylinder:



$\rightarrow E_{in} = \frac{\rho r}{2\varepsilon_0}$

Metal

① Solid sphere:



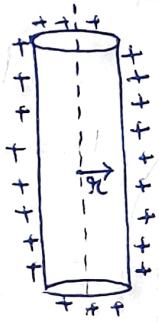
$\rightarrow E_{in} = 0$

$\rightarrow V_{in} = \frac{KQ}{R} = V_c$

$\rightarrow U_{self} = \frac{KQ^2}{2R}$

∴ It behaves like hollow sphere

② Infinitely long solid cylinder:



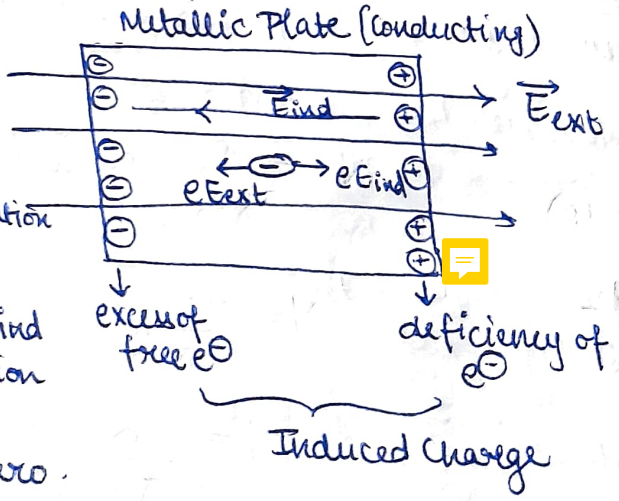
$\rightarrow E_{in} = 0$

→ Electric field inside bulk of conductor will be zero.

As soon as $\vec{E}_{ext}$ is applied -ve charge and +ve charge gets induced due to movement of $e^{\ominus}$.

As -ve charge accumulation increases, $|\vec{E}_{ind}| \uparrow$.

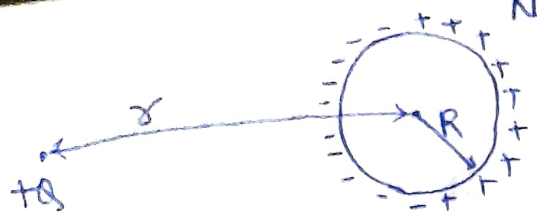
As soon as $E_{ext} = E_{ind}$ electrostatic condition is achieved and $\vec{E}_{net}$ inside is zero.



$\vec{E}_{ind} = -\vec{E}_{ext}$

$\therefore \vec{E}_{net} = 0$

Ex



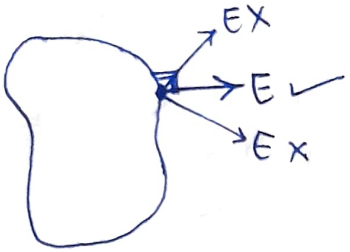
Neutral conductor

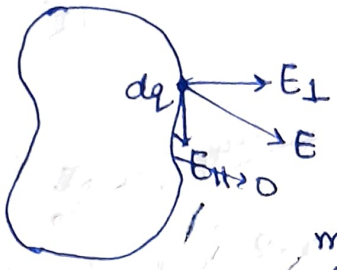
Find ① E·F at point C
② E·F at point C due to +Q
③ E·F at point C due to induced charge

① $E_c = 0$ ② $\frac{KQ}{r^2}(\rightarrow)$ ③ $E_{net} = 0$, $E_{+Q} = \frac{KQ}{r^2}$, E_{ind} — C — E_{+Q} $\therefore E_{ind} = \frac{KQ}{r^2}(\leftarrow)$

→ ELOF will never enter into conductor

→ Electric field at the surface of conductor is always ⊥ to the surface

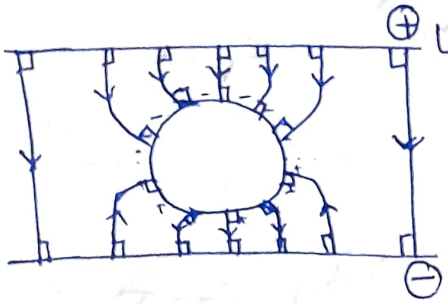




If E·F is not ⊥ to surface then its parallel component along surface will make it move along surface of conductor and will break the electrostatic cond^n.

→ ELOF will strike the conductor normally.

Q



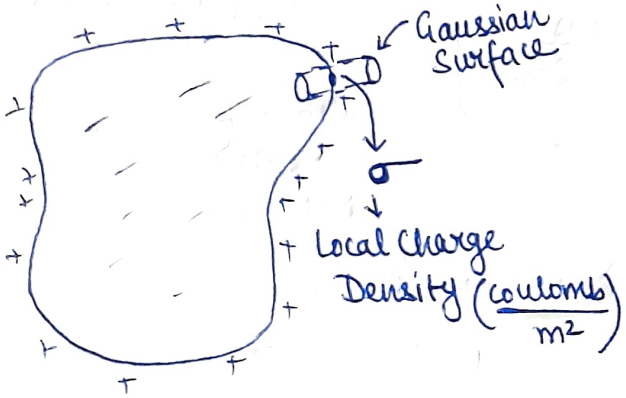
Large Mettalic Plate

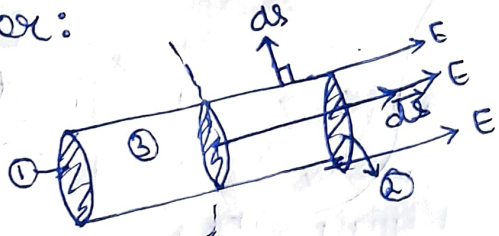
Mettalic Sphere (neutral)

Large Mettalic Plate

Draw ELOF in the region

Electric Field at the surface of conductor:





By definition $\phi = \phi_1 + \phi_2 + \phi_3$

$\phi = 0 + E\,ds + 0$

$\phi = E\,ds$ — ①

By Gauss law: $\phi = \frac{Q_{in}}{\varepsilon_0}$

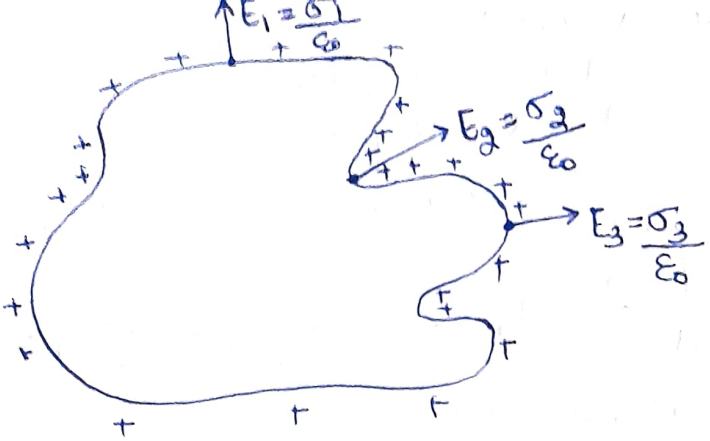
$\phi = \frac{\sigma\,ds}{\varepsilon_0}$ — ②

Now, ① = ②

$E\,ds = \frac{\sigma\,ds}{\varepsilon_0}$

$\therefore$ Electric field at surface of conductor $= E = \frac{\sigma}{\varepsilon_0}$

σ → Local Charge density

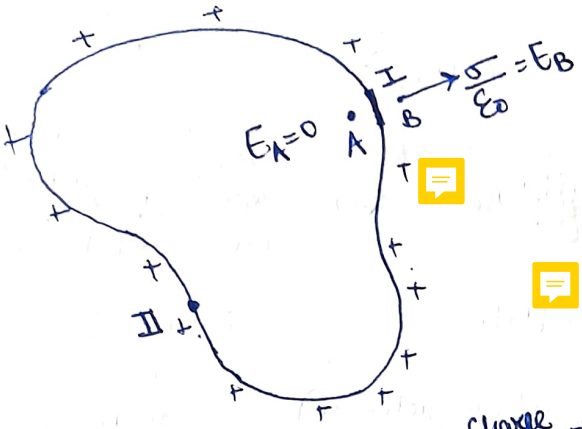


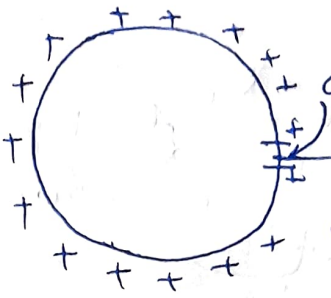
★ $\sigma \propto \frac{1}{\text{Radius of curvature}}$

If surface is flat ⇒ R↑ ⇒ σ↓
If surface is sharp ⇒ R↓ ⇒ σ↑

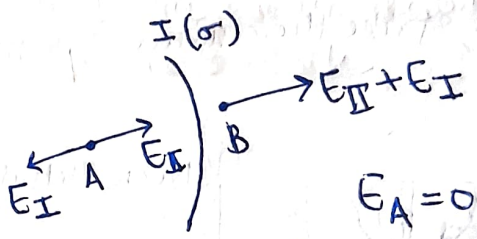
Electric Pressure : $P = \frac{dF}{dS}$

* Derivation :





↓ Force due to other charges on this elemental charge



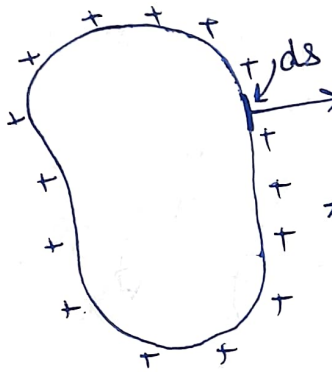
$E_A = 0$

$\Rightarrow E_I = E_{II}$ — (1)

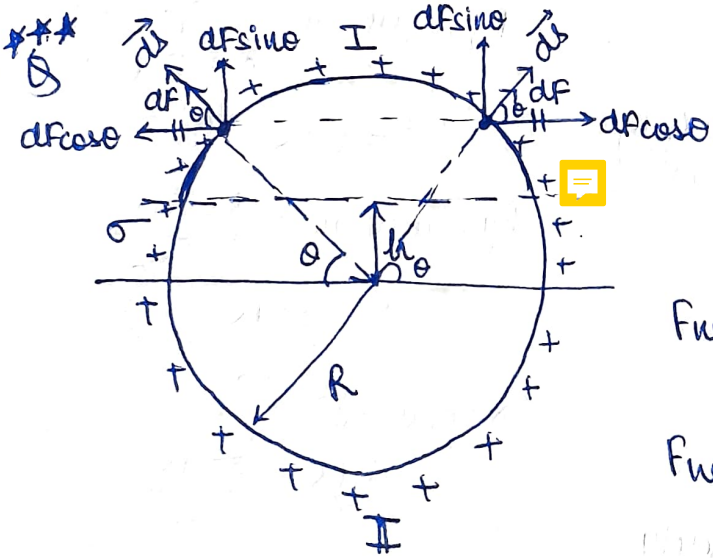
$E_I + E_{II} = \frac{\sigma}{\varepsilon_0}$ — (2)

$\therefore \boxed{E_I = E_{II} = \frac{\sigma}{2\varepsilon_0}}$ ★

↳ E·F due to charge on which electric pressure is to be found is $\sigma/2\varepsilon_0$
Also, E·F. due to other charges is also $\sigma/2\varepsilon_0$.
Net Sum = $\frac{\sigma}{\varepsilon_0}$.



★ $\boxed{P = \frac{dF}{dS} = \frac{\sigma^2}{2\varepsilon_0}}$

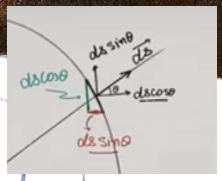


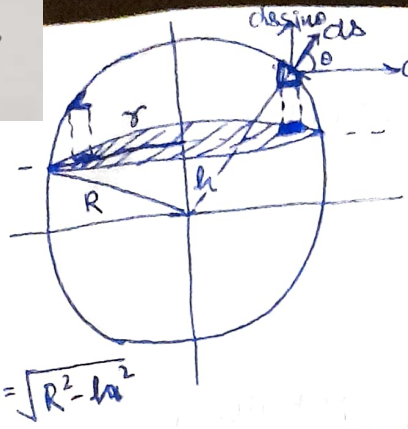
Find electrostatic force on part I

$dF = P\,ds = \frac{\sigma^2}{2\varepsilon_0} ds$

$F_{net} = \int dF \sin\theta = \int \frac{\sigma^2}{2\varepsilon_0} ds (\sin\theta)$

$F_{net} = \frac{\sigma^2}{2\varepsilon_0} \int ds \sin\theta$ — (1)





$r = \sqrt{R^2 - h^2}$

* $ds\sin\theta$ will projection of the surface for which $ds\sin\theta$ acts as area vector or $\perp^{rd}$ surface of $\vec{ds}$

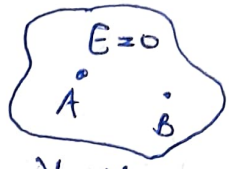
$$F_{net} = \oint \frac{\sigma^2}{2\varepsilon_0} \int (ds\sin\theta) = \frac{\sigma^2}{2\varepsilon_0}(S_\perp)$$

$$F_{net} = \frac{\sigma^2}{2\varepsilon_0}\pi r^2 = \frac{\sigma^2}{2\varepsilon_0}\pi(R^2 - h^2)$$

$$\therefore \boxed{F_{net} = \frac{\sigma^2}{2\varepsilon_0}\pi(R^2 - h^2)}$$

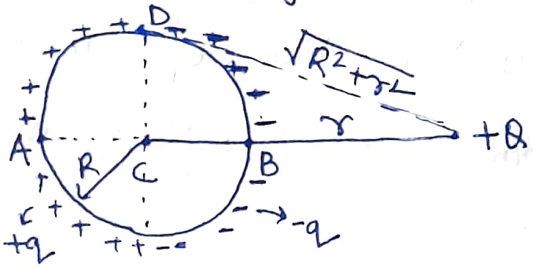
Properties of Conductor (continued):

→ Conductor has equal electric potential at any point on it. OR Conductor is an equipotential surface.



$V_A = V_B$

Q Neutral Conducting sphere



① Find electric potential of the sphere

$$V_C = \frac{KQ}{r} + \frac{Kq}{R} + \frac{K(-q)}{R}$$

$$\boxed{V_C = \frac{KQ}{r}} \qquad \boxed{V_C = V_A = V_B = V_D}$$

② Find V due to induced charge at

ⓐ Point A: $V_A = \frac{KQ}{r} = V_{+Q} + V_{ind}$

$$\frac{KQ}{r} = \frac{KQ}{R+r} + V_{ind}$$

$$\boxed{V_{ind} = \frac{KQ}{r} - \frac{KQ}{R+r}}$$

ⓑ Point B: $V_B = \frac{KQ}{r} = V_{+Q} + V_{ind}$

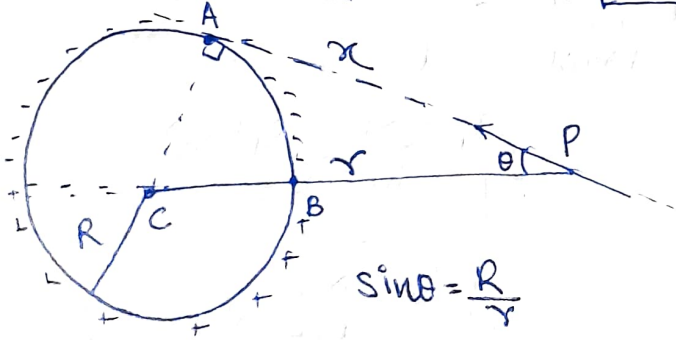
$$\frac{KQ}{r} = \frac{KQ}{R-R} + V_{ind}$$

$$\boxed{V_{ind} = \frac{KQ}{r} - \frac{KQ}{r-R}}$$

ⓒ Point D: $V_D = \frac{KQ}{r} = V_{+Q} + V_{ind}$

$$\boxed{V_{ind} = \frac{KQ}{r} - \frac{KQ}{\sqrt{R^2+r^2}}}$$

Q Conducting neutral sphere



$\sin\theta = \frac{R}{r}$

$\tan\theta = \frac{R}{x}$

① Find V of conducting sphere

$$V_{sphere} = V_C = \frac{KP\cos\theta}{r^2}$$

$$\boxed{V_C = \frac{KP\cos\theta\sin^2\theta}{R^2}}$$

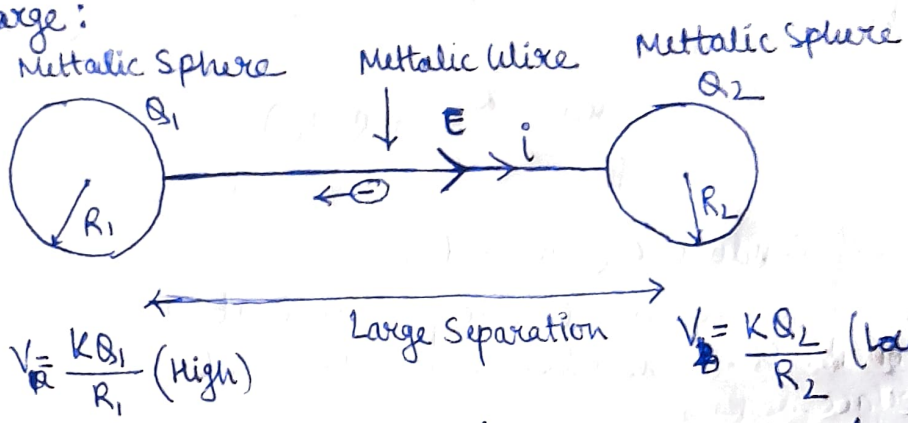
$$\boxed{V_A = V_B = V_C}$$

② Find electric potential at point A due to induced charge

$$V_A = \frac{KP\cos\theta\sin^2\theta}{R^2} = V_p + V_{ind} = \frac{KP\tan^2\theta}{R^2} + V_{ind}$$

$$\boxed{V_{ind} = \frac{KP\cos\theta\sin^2\theta}{R^2} - \frac{KP\tan^2\theta}{R^2}}$$

Sharing of Charge:



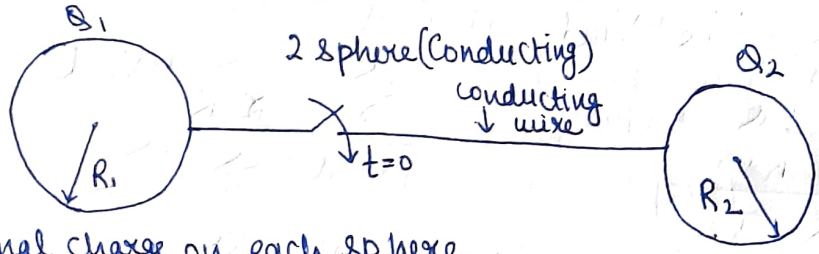
As current starts flowing, Q_1 charge ↓ (due to accumulation of more $e^\ominus$) due to which V_a ↓. Similarly Q_2 ↑ due to which V_b ↑.
Let final charges be $Q_1 \to q_1$ and $Q_2 \to q_2$. Also let final potentials be V_1 and V_2.

★ Charge will flow from High V to lower V till their final potentials get equal.

$\therefore$ Finally $\boxed{V_1 = V_2}$

★ Charge Conservation ⇒ $\boxed{Q_1 + Q_2 = q_1 + q_2}$

Q



ⓐ Find final charge on each sphere.
Let final potential be V_1 and V_2. Also, $Q_1 \to q_1$ $\quad Q_2 \to q_2$

$V_1 = V_2$

$$\frac{Kq_1}{R_1} = \frac{Kq_2}{R_2}$$

$$\boxed{\frac{q_1}{q_2} = \frac{R_1}{R_2}} \quad -①$$

In this case, spheres will share charge directly proportional to radius.

C/C ⇒ $Q_1 + Q_2 = q_1 + q_2$ $\quad -②$

From ① and ②

$$\boxed{q_1 = \frac{R_1}{R_1 + R_2}(Q_1 + Q_2)}$$

$$\boxed{q_2 = \frac{R_2}{R_1 + R_2}(Q_1 + Q_2)}$$

(b) Find final charge per unit area on each sphere.

$$\sigma_1 = \frac{q_1}{4\pi R_1^2} = \frac{(Q_1+Q_2)}{4\pi R_1(R_1+R_2)}$$

$$\sigma_2 = \frac{q_2}{4\pi R_2^2} = \frac{(Q_1+Q_2)}{4\pi R_2(R_1+R_2)}$$

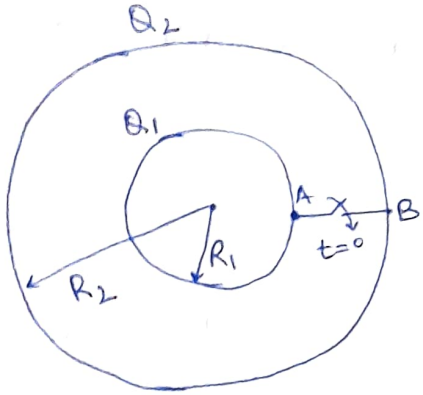
$$\boxed{\frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}} \qquad \boxed{\sigma \propto \frac{1}{R}}$$

(c) Find energy lost from system.
Energy will be lost due to heat dissipated.

Heat = Electric Energy lost = $U_i - U_f$

$$= \left(\frac{KQ_1^2}{2R_1} + \frac{KQ_2^2}{2R_2}\right) - \left(\frac{Kq_1^2}{2R_1} + \frac{Kq_2^2}{2R_2}\right)$$

Q



2 conducting Concentric shell

(a) Find final charge on each shell.
Let final charge on inner shell = q
Final charge on outer shell = Q_1+Q_2-q

Finally, $V_A = V_B$

$$\frac{Kq}{R_1} + \frac{K(Q_1+Q_2-q)}{R_2} = \frac{Kq}{R_2} + \frac{K(Q_1+Q_2-q)}{R_2}$$

$$\frac{Kq}{R_1} - \frac{Kq}{R_2} = 0 \Rightarrow Kq\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = 0$$

$\therefore \boxed{q = 0}$ Final charge on inner shell = 0

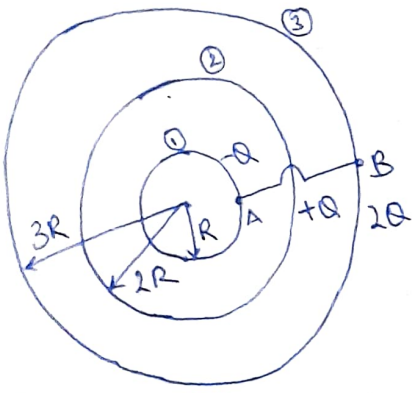
Final charge on outer shell = Q_1+Q_2

(b) Find heat lost in process

Heat lost = $U_i - U_f$

$$= \left(\frac{KQ_1^2}{2R_1} + \frac{KQ_2^2}{2R_2} + \frac{KQ_1Q_2}{R_2}\right) - \left(\frac{K(Q_1+Q_2)^2}{2R_2}\right)$$

Q



3 concentric conducting shells

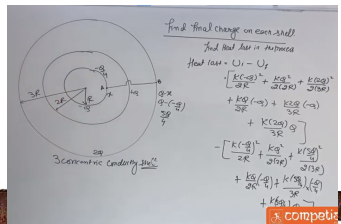
Find final charge on each shell.
Let final charge on ① be x
final charge on ③ be $2Q - Q - x = Q - x$
Now, $V_A = V_B$

$$= \frac{Kx}{R} + \frac{KQ}{2R} + \frac{K(Q-x)}{3R} = \frac{Kx}{3R} + \frac{KQ}{3R} + \frac{K(Q-x)}{3R}$$

$$= x + \frac{Q}{2} = \frac{x}{3} + \frac{Q}{3}$$

$$= \frac{2x}{3} = \frac{Q}{3} - \frac{Q}{2} = -\frac{Q}{6}$$

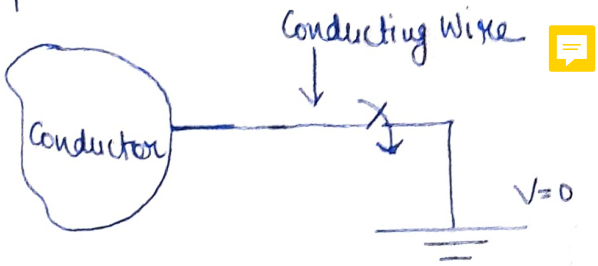
$\boxed{x = -\frac{Q}{4}}$ Final charge on ① = $-Q/4$
Final charge on ③ = $5Q/4$



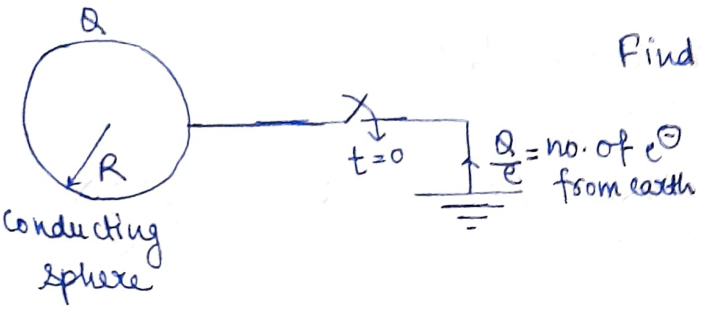
Earthing / Grounding :

→ Electric Potential of earth is assumed to be zero and constant.

→ Whenever we connect a conducting body to earth through a conducting wire its potential will get same as earth ie zero.



Q

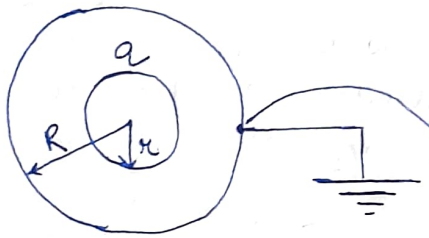


Find final charge on shell.

Let final charge on sphere be x.

$$V = \frac{Kx}{R} = 0 \Rightarrow \boxed{x = 0}$$

Q

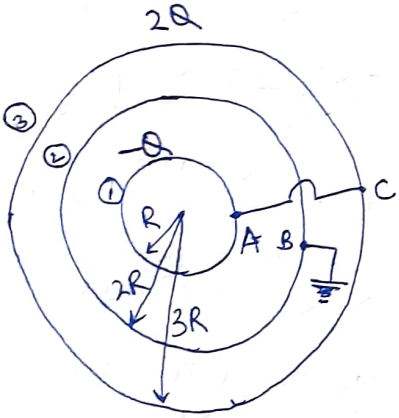


Find charge on outer shell.

Let charge on outer shell be x

$$V = \frac{Kq}{R} + \frac{Kx}{R} = 0 \Rightarrow \boxed{x = -q}$$

**
Q



Find final charge on each shell.

Let final charge on ① be x

final charge on ③ be $2Q - Q - x = Q - x$

final charge on ② be y.

$$V_A = V_C$$

$$= \frac{Kx}{R} + \frac{Ky}{2R} + \frac{K(Q-x)}{3R} = \frac{Kx}{3R} + \frac{Ky}{3R} + \frac{K(Q-x)}{3R}$$

$$\Rightarrow x + \frac{y}{2} = \frac{x}{3} + \frac{y}{3}$$

$$\boxed{x = \frac{-y}{4}} \quad \text{—①}$$

Also, $V_B = 0$

$$\frac{Kx}{2R} + \frac{Ky}{2R} + \frac{K(Q-x)}{3R} = 0$$

$$= \frac{x}{2} + \frac{y}{2} + \frac{Q-x}{3} = 0$$

$$= \frac{x}{2} - 2x + \frac{Q}{3} - \frac{x}{3} = 0 \quad \left\{\frac{y}{2} = -2x \text{ from ①}\right\}$$

$$= \frac{x}{6} - 2x + \frac{Q}{3} = 0$$

$$= 2x - \frac{x}{6} = \frac{Q}{3}$$

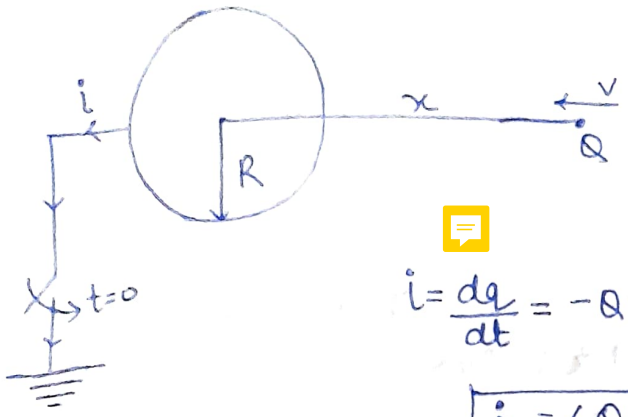
$$\boxed{x = \frac{2Q}{11}}$$

∴ Final charge on sphere ① $= \frac{2Q}{11}$

Final charge on sphere ② $= \frac{-8Q}{11}$

Final charge on sphere ③ $= \frac{9Q}{11}$

Q Conducting sphere



Find current in wire connecting sphere and earth

Let final charge on sphere be q

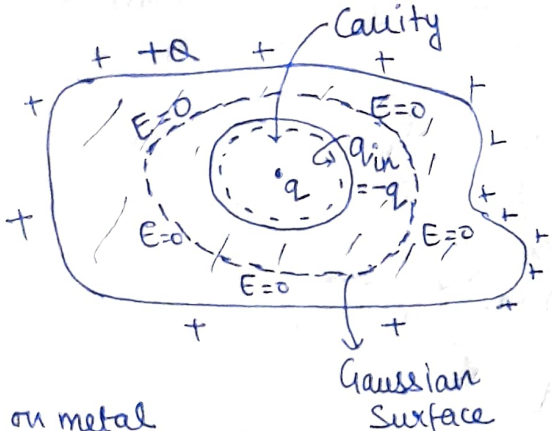
$$V = \frac{Kq}{R} + \frac{KQ}{x} = 0$$

$$\boxed{q = -\frac{QR}{x}}$$ As $x \downarrow$ $|q| \uparrow$

$$i = \frac{dq}{dt} = -QR\left\{-\frac{1}{x^2}\frac{dx}{dt}\right\}$$

$$\boxed{i = \left(\frac{QR}{x^2}\right)v}$$

Charge distribution and induction in case of cavity:



By definition: $\phi = \oint \vec{E} \cdot \vec{ds}$

$\phi = 0$ — (1) $\{\vec{E} = 0\}$

By gauss law: $\phi = \frac{Q_{in}}{\varepsilon_0}$

$\phi = \frac{q + q_{in}}{\varepsilon_0}$ — (2)

From (1) and (2)

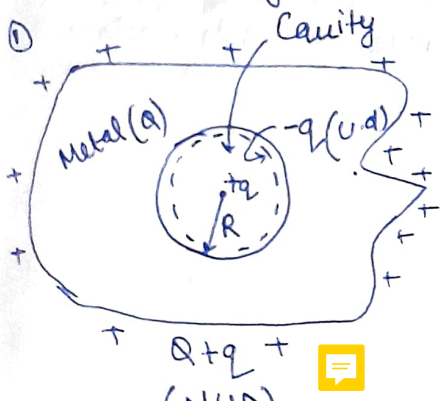
$q + q_{in} = 0$

$$\boxed{q_{in} = -q}$$

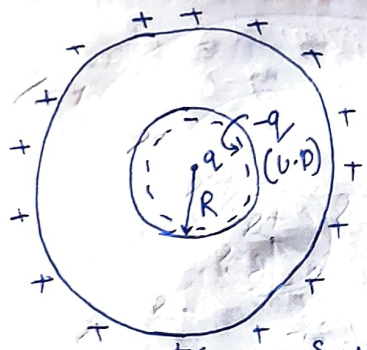
Charge on metal after induction $= Q + q$

Examples of charge distribution in different cases:

(1)

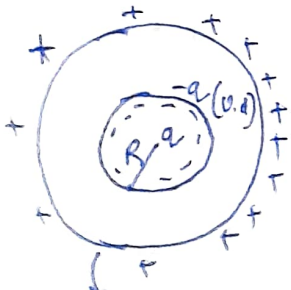


(2)

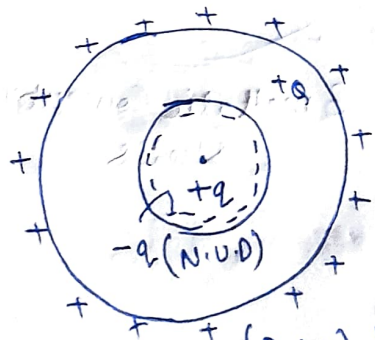


Metallic sphere with concentric cavity.

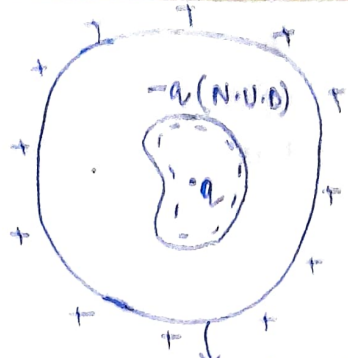
(3)



(4)

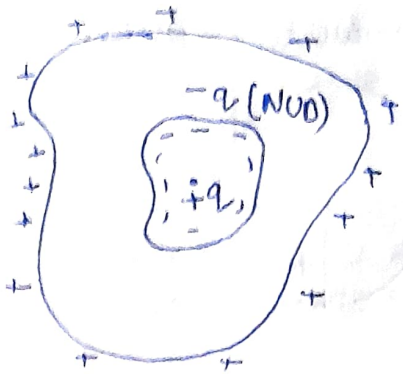


(5)



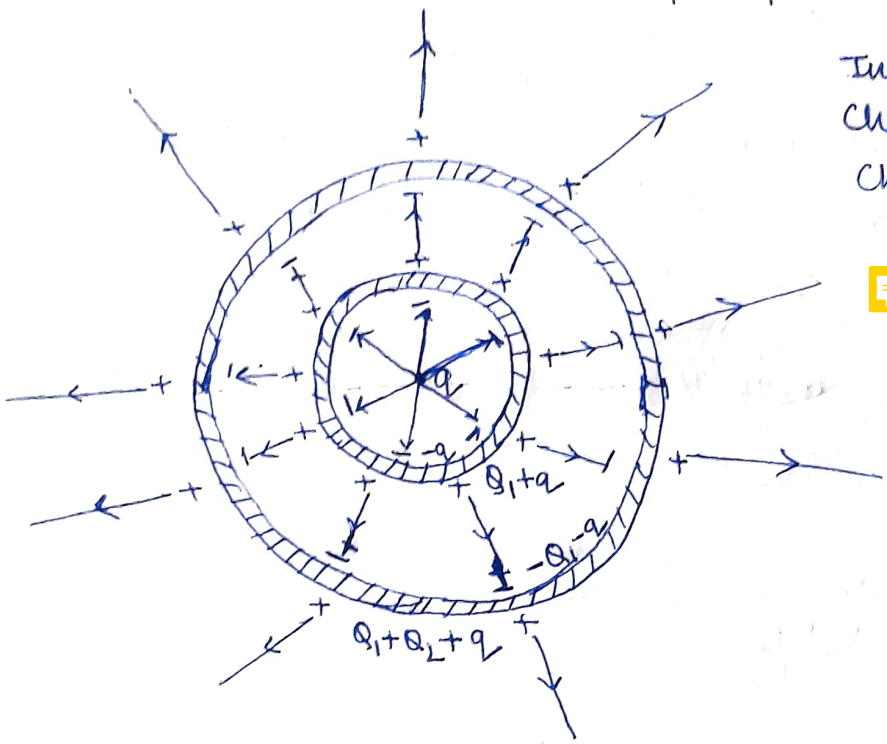
Q+q {U.D}

(6)



Q+q {NUD}

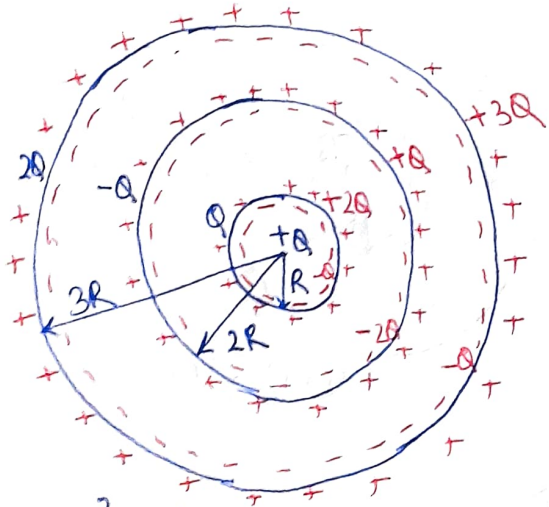
Charge distribution on inner and outer surface of concentric mettalic shells:



Initially:
Charge on ① = Q_1
Charge on ② = Q_2 →

Q

Q



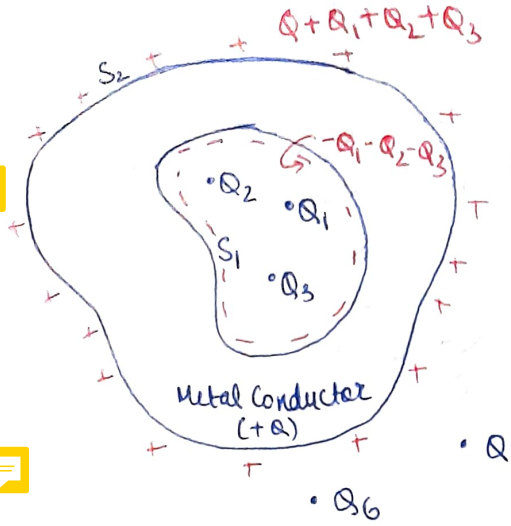
3 mettalic Concentric Shells

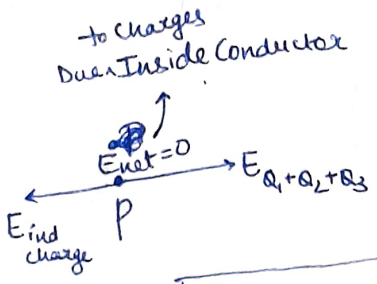
Find Charge on each surface of shell.

Original Charges marked with ble Q

Induced charges marked with red

Electrostatic shielding:

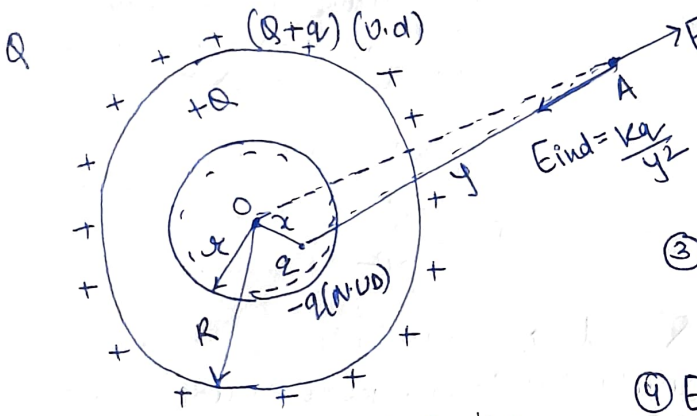




★ Charge distribution on S_1 ⇒
i) depends on Q_1, Q_2 and Q_3
ii) depends on surface S_1

★ Charge distribution on S_2 ⇒
i) depends on outer charges Q_4, Q_5, Q_6
ii) depends on outer surface S_2

→ Charges inside S_1 and charge induced on S_1 will have no net electric field outside S_1.
Charges outside S_2 and charge induced on S_2 will have no net electric field inside S_2.
This phenomenon is known as electrostatic shielding.



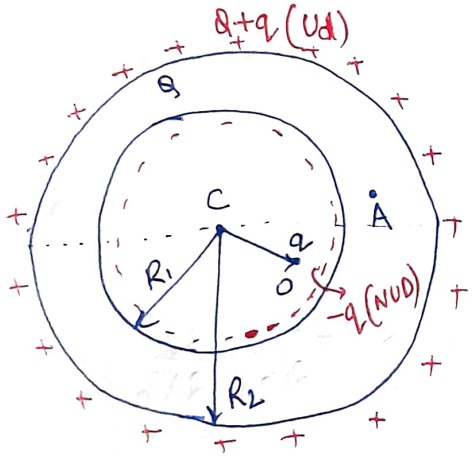
Metallic sphere with cavity

Find
① $E_A = \frac{K(Q+q)}{(OA)^2}$

② E_A due to $q = \frac{Kq}{y^2}$

③ E_A due to charge induced on inner surface of sphere = $\frac{Kq}{y^2}$

④ E_A due to charge on outer surface of sphere = $\frac{K(Q+q)}{(OA)^2}$



Spherical metallic conductor with spherical cavity.

Find ① V_C ② V_A ③ V_B

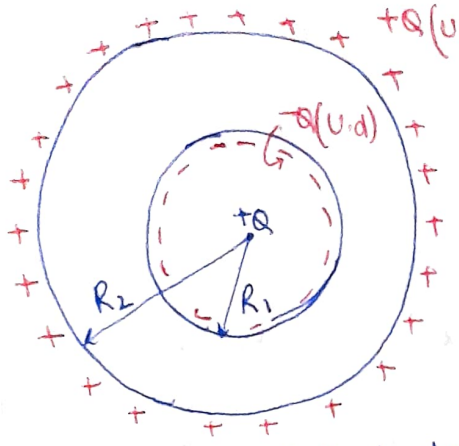
① $V_C = \frac{Kq}{(OC)} + \frac{K(-q)}{R_1} + \frac{K(Q+q)}{R_2}$

② $V_A = \frac{K(Q+q)}{R_2}$

③ $V_B = \frac{K(Q+q)}{OB}$

+q and -q will have no electric field outside shell of radius R_1, so E due to them will be zero and they will have no contribution in Wext to bring charge from ∞ to A

Q



Neutral Spherical Conductor

Find work done by external agent to bring +Q charge from ∞ to center of sphere.

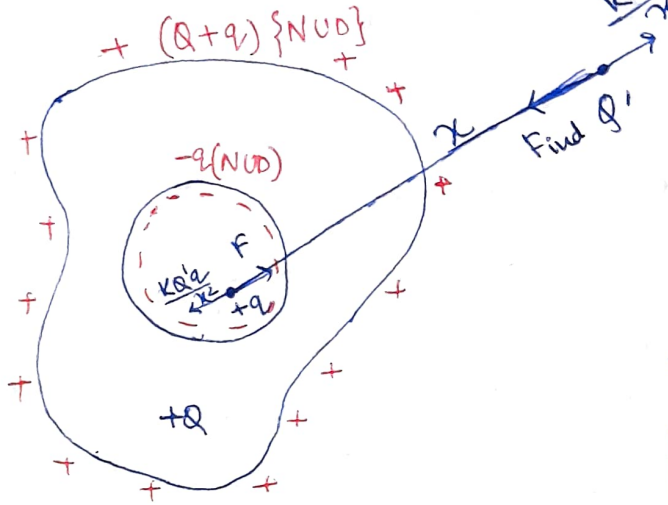
$U_i = 0$

$$U_f = \frac{K(-Q)^2}{2R_1} + \frac{KQ^2}{2R_2} + \frac{K(-Q)(Q)}{R_1} + \frac{KQQ}{R_2} + \frac{K(-Q)(Q)}{R_2}$$

$$U_f = \frac{KQ^2}{2R_2} - \frac{KQ^2}{2R_1}$$

$$\boxed{W_{ext} = U_f - U_i = \frac{KQ^2}{2R_2} - \frac{KQ^2}{2R_1}}$$

Q



① Find electrostatic force applied by charge on outer surface on +q

We know that Enet inside the cavity due to outer charge and charge on outer surface must be zero ⇒ Enet = 0

⇒ $F_{Q'q} = F_q$ (charge on outer surface)

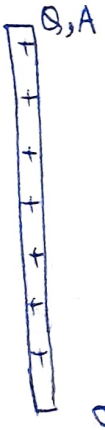
$$\boxed{F = \frac{KQ'q}{x^2}}$$

② Find electrostatic force b/w Q' and −q charge induced on inner surface. Similarly Enet outside the cavity due to charges in cavity will be zero

$$\boxed{F = \frac{KQ'q}{x^2}}$$

Metallic / Conducting Large sheet:

In case of Non-conducting sheet:



$\leftarrow E = \frac{\sigma}{2\varepsilon_0} = \frac{Q}{2A\varepsilon_0}$

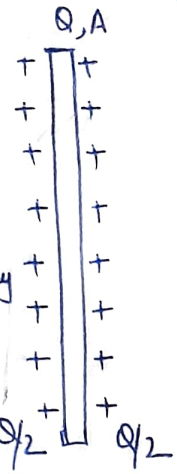
$\rightarrow E = \frac{\sigma}{2\varepsilon_0} = \frac{Q}{2A\varepsilon_0}$

$\sigma = \frac{Q}{A}$

Conducting isolated sheet:

For conducting sheet charge is spread on surface. For uniform distribution we can say charge Q/2 is distributed on each face of area A. So effectively charge Q is spread on area 2A.

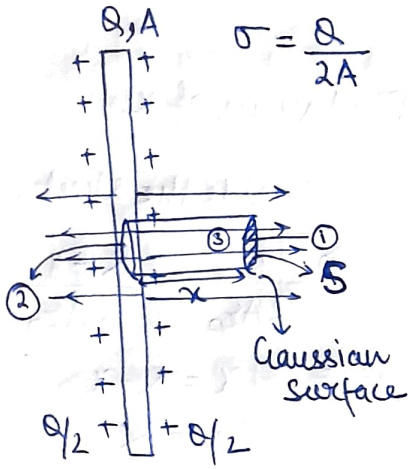
$\sigma = \frac{Q/2}{A} = \frac{Q}{2A}$



$\longrightarrow \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0} = \frac{Q}{2A\varepsilon_0}$

Also, $\boxed{\sigma_{N.C\,sheet} = 2\sigma_{C\,sheet}}$

Electric Field due to conducting large sheet using gauss law:



$\sigma = \frac{Q}{2A}$

By definition, $\phi = \phi_1 + \phi_2 + \phi_3$

$= ES\cos 0° + 0 + 0$

$\phi = ES$ — (1)

By Gauss law: $\phi = \frac{Q_{in}}{\varepsilon_0} = \frac{\sigma S}{\varepsilon_0}$ — (2)

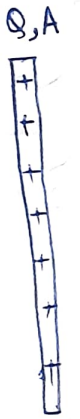
Now (1) = (2) $\Rightarrow$ $ES = \frac{\sigma S}{\varepsilon_0}$

$\boxed{E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{2A\varepsilon_0}}$

Comparison between conducting and non-conducting large sheets:

① Charge Same

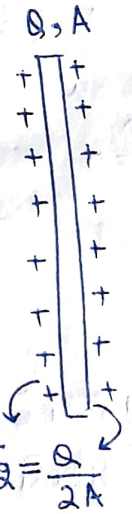
Non-Conducting



$\sigma_1 = \frac{Q}{A}$

$\longrightarrow E_1 = \frac{Q}{2A\varepsilon_0} = \frac{\sigma_1}{2\varepsilon_0}$

Conducting



$\sigma_2 = \frac{Q}{2A}$

$\longrightarrow E_2 = \frac{Q}{2A\varepsilon_0} = \frac{\sigma_2}{\varepsilon_0}$

$\boxed{E_1 = E_2}$

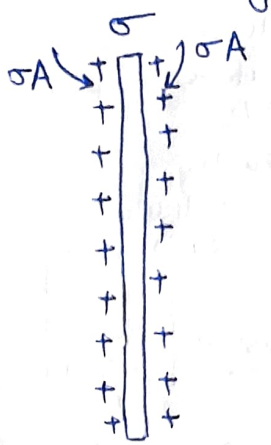
② Same σ

Non-Conducting



$Q_1 = \sigma A$ ⟶ $E_1 = \frac{\sigma}{2\varepsilon_0} = \frac{Q_1}{2A\varepsilon_0}$

Conducting



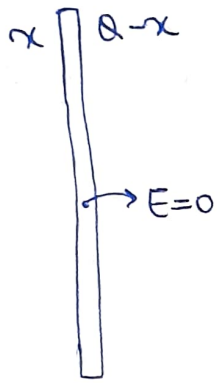
$Q_2 = 2\sigma A$

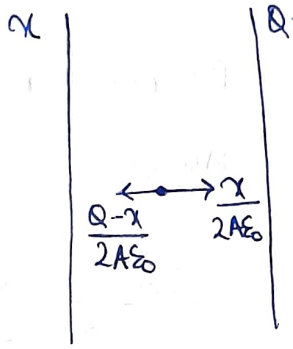
$\boxed{Q_2 = 2Q_1}$

⟶ $E_2 = \frac{\sigma}{\varepsilon_0} = \frac{Q_2}{2A\varepsilon_0}$

$\boxed{E_2 = 2E_1}$

Q Isolated large conducting sheet Find value of x.





Inside the sheet

Inside the sheet, E=0

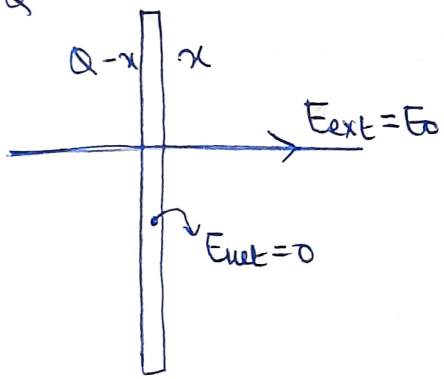
$\Rightarrow \frac{x}{2A\varepsilon_0} = \frac{Q-x}{2A\varepsilon_0}$

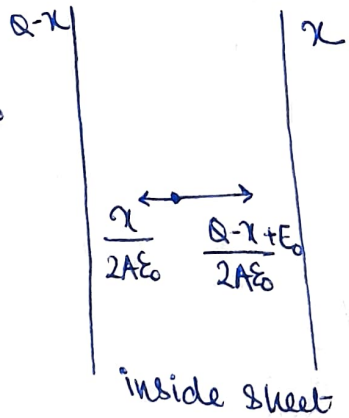
$= x = Q - x$

$2x = Q$

$\boxed{x = Q/2}$

Q





inside sheet

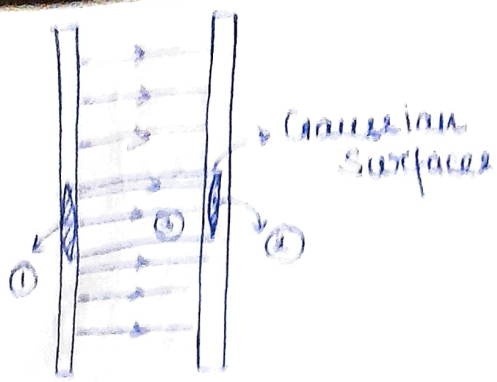
Find value of x

$\Rightarrow \frac{x}{2A\varepsilon_0} = \frac{Q-x}{2A\varepsilon_0} + E_0$

$= x = Q - x + 2A\varepsilon_0 E_0$

$= \boxed{x = \frac{Q}{2} + A\varepsilon_0 E_0}$

* Note: Two metallic plates are placed parallely, then the faces of plate opposing each other will have equal and opposite charge.



2 metallic Plates

By definition: $\phi = \phi_1 + \phi_2 + \phi_3$
$= 0 + 0 + 0$

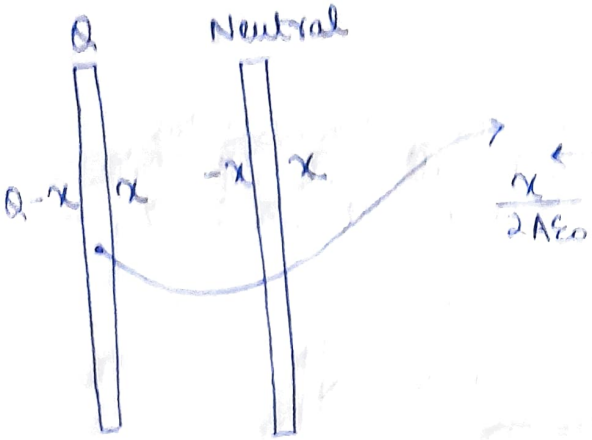
$$\boxed{\phi = 0} \quad -(1)$$

By Gauss law: $\phi = \frac{Q_{in}}{\varepsilon_0}$ $-(2)$

$$\therefore \boxed{Q_{in} = 0}$$

∴ Both faces should have charge equal and opposite.

Q



Two metallic large sheets.

Find charge on each face.

$\frac{x}{2A\varepsilon_0}$ ← $E=0$ → $\frac{Q-x}{2A\varepsilon_0}$

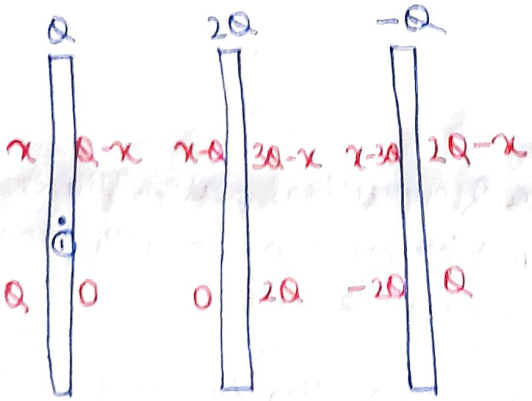
{ Electric field due to $x, -x$ on inner side will be equal and opp and cancel }

$\Rightarrow x = Q - x$

$$\boxed{x = Q/2}$$

[Trick: Leftmost E ≡ Rightmost E]

Q



3 large metallic plates

Find charge on each face.

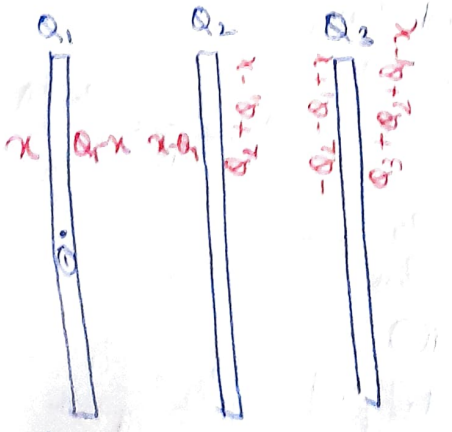
At point ① $\Rightarrow E = 0$

$\frac{2Q-x}{2A\varepsilon_0}$ ← $E=0$ → $\frac{x}{2A\varepsilon_0}$

$\Rightarrow 2Q - x = x$

$$\boxed{x = Q}$$

Q



3 large metallic plates

Find charge on each face

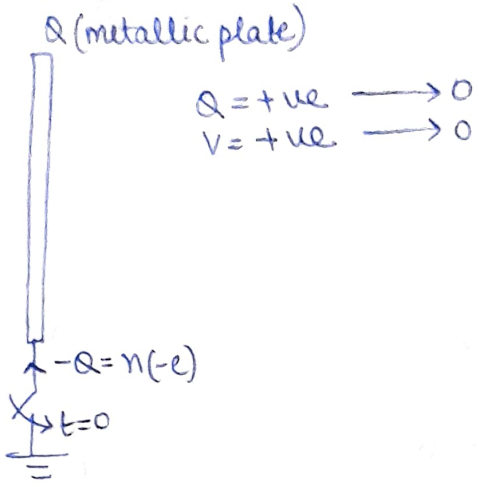
At point ① $\Rightarrow E = 0$

$\frac{Q_1+Q_2+Q_3-x}{2A\varepsilon_0}$ ← $E=0$ → $\frac{x}{2A\varepsilon_0}$

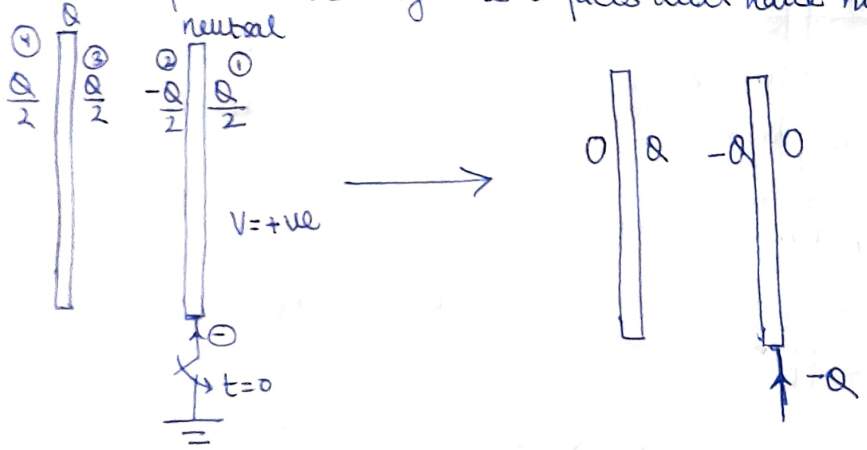
$Q_1 + Q_2 + Q_3 - x = x$

$$\boxed{x = \frac{Q_1 + Q_2 + Q_3}{2}}$$

Earthing of Metallic Plate :

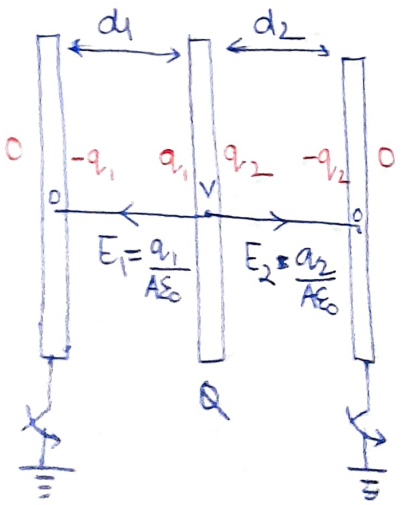


* If we have two plates and we earth one of the metallic plate, then earth will provide charge such that final V of plate becomes zero and leftmost and rightmost faces will have no charge.



Initially V = +ve due to plate of charge Q on neutral plate. We know due to earthing V of neutral plate must become zero. So earth provides negative charge. This -ve charge increases ② due to which ③ also increases. Also ① ↓ and similarly ④ also ↓. This process goes on until charge on on ① and ④ becomes zero. Also V at neutral plate is now zero.

Q*



① Find Charge on each face.

$q_1 + q_2 = Q$ — ①

$V = E_1 d_1 = E_2 d_2 \Rightarrow \frac{E_1}{E_2} = \frac{d_2}{d_1} = \frac{q_1/A\varepsilon_0}{q_2/A\varepsilon_0}$

$\frac{d_2}{d_1} = \frac{q_1}{q_2}$ — ②

From ① and ②

$$q_1 = \frac{d_2}{d_1 + d_2} Q$$

$$q_2 = \frac{d_1}{d_1 + d_2} Q$$

② Find V of the plate in middle.

$V = E_1 d_1 = E_2 d_2$

$V = \frac{q_1}{A\varepsilon_0} d_1 = \frac{Q d_1 d_2}{A\varepsilon_0 (d_1 + d_2)}$

$$\boxed{V = \frac{Q d_1 d_2}{A\varepsilon_0 (d_1 + d_2)}}$$

Dielectrics:

→ It is a medium which doesn't have many free electrons like conductors.

→ Charge is mostly bounded.

a) Polar Dielectric:

→ In molecule, +ve charge and −ve charge centres doesn't coincide.

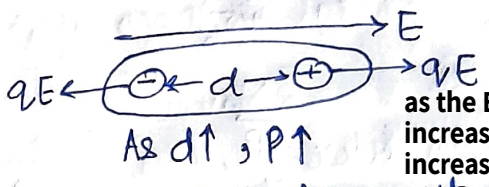
→ will have some dipole moment

→ They have ionic bond.

→ Ex - NaCl, HCl

→ They can also form covalent polar bonds.





as the EF is increased, d will increase

b) Non-polar Dielectric:

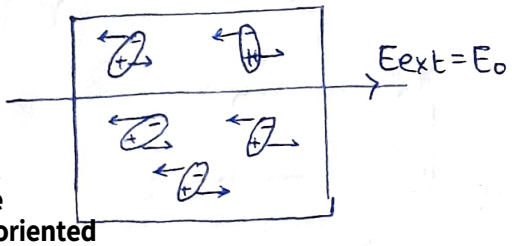
→ In molecule, +ve and −ve charge centres coincide.

→ If Eext is applied, it creates a dipole moment.

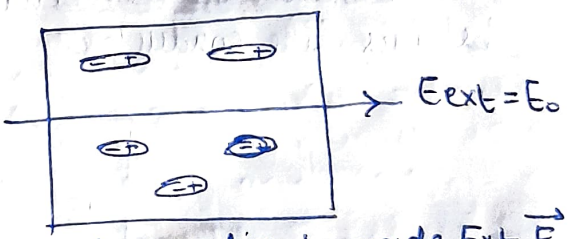




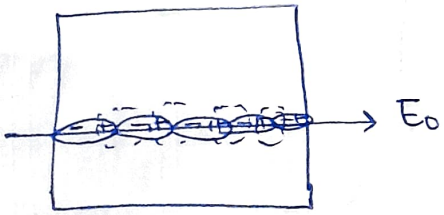
Polarization:



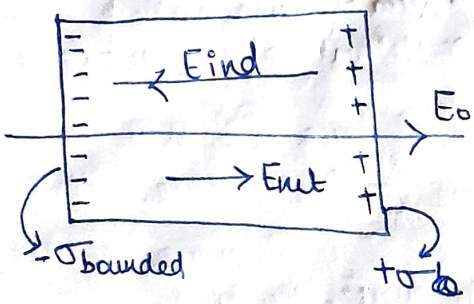
initially, all dipoles are randomly oriented so the net dipole moment will be zero



Molecules align towards Ext $\vec{E}$



All charges neutralise except liftmost and right most charge



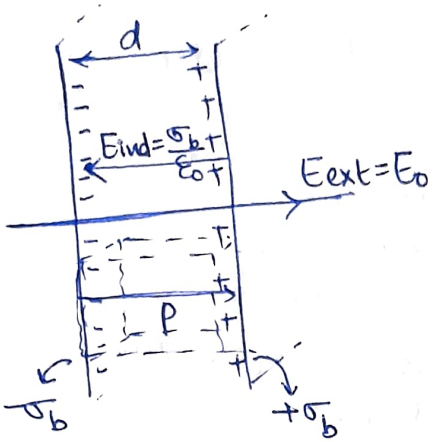
$$\boxed{E_{net} = E_0 - E_{ind} = \frac{E_0}{\varepsilon_r}}$$

ε_r → Relative Permitivity, dielectric constant

$$E_{net} = E_0 - E_{ind} = \frac{E_0}{\varepsilon_r} \Rightarrow E_{ind} = E_0 - E_{net}$$

$$\boxed{E_{ind} = E_0 - \frac{E_0}{\varepsilon_r} = E_0\left\{1 - \frac{1}{\varepsilon_r}\right\}}$$

⊛ Large Metallic Plate



$$E_{net} = E_0 - E_{ind} = E_0 - \frac{\sigma_b}{\varepsilon_0} = \frac{E_0}{\varepsilon_r}$$

$$E_0\left\{1 - \frac{1}{\varepsilon_r}\right\} = \frac{\sigma_b}{\varepsilon_0}$$

$$\star \boxed{\sigma_b = \varepsilon_0 E_0\left\{1 - \frac{1}{\varepsilon_r}\right\}}$$

$$\boxed{\text{Polarization Vector} = \frac{\vec{P}}{\text{Volume}}}$$

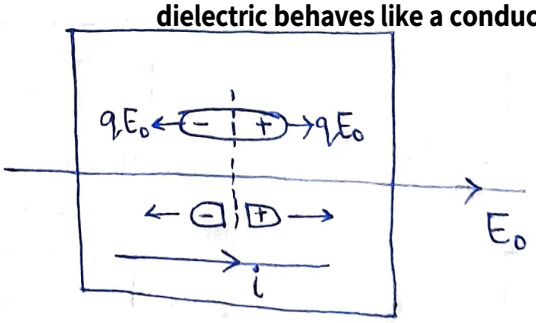
- → Vector
- → Direcⁿ along $\vec{P}$
- → Unit = C/m^2, Dimension = $[I\,T\,L^{-2}]$

$$\text{Polarization Vector} = \frac{\vec{P}}{\text{Volume}} = \frac{(\sigma_b A)d}{Ad} = \sigma_b \hat{n}$$

$$\therefore \boxed{\text{Polarization Vector} = \varepsilon_0 \vec{E_0}\left\{1 - \frac{1}{\varepsilon_r}\right\}}$$

Dielectric Strength:

→ It is minimum electric field required to break molecules and these broken molecules behave like charge carriers and dielectric behaves like conductor. **When the dipoles break, suddenly a large no. of charge carriers are created due to which the current shoots up suddenly and the dielectric behaves like a conductor**



Irodov Questions: 3.1 to 3.39